

Technical Architecture

The solution follows a layered architecture consisting of Data, Machine Learning, API, and Presentation layers.

The Data Layer handles raw property datasets, preprocessing, cleaning, and storage. The Machine Learning Layer performs feature engineering, categorical encoding, model training, and prediction generation using XGBoost.

The API Layer is implemented using FastAPI and acts as the communication bridge between the frontend and machine learning model. It receives user input, transforms data, invokes the model, and returns predictions.

The Presentation Layer uses Streamlit to provide a user-friendly interface for data entry and result visualization. This separation of concerns improves maintainability, scalability, and future extensibility of the application.